

CSCI 131 — Introduction to Programming in Python

Programming Assignment 4

Nested Conditionals and Logical Operators

Problem 1 has no starter files. Open IDLE, choose **File** → **New File** and save it under `login.py`. Put your name in a comment on the first line.

Problem 2 has a starter file, `rps.py`, (on Canvas or the course website). Download and open it with IDLE, **File** → **Open**

Problem 1 — Login

Write a program `login.py` which asks for a username and a PIN, then checks them against two accounts. The user `alice` has PIN `1234`, and the user `bob` has PIN `9876`. Use the prompts `'username = input('Username: ')` and `'pin = input('PIN: ')`, in that order.

If the username is not one of the two accounts, print `Unknown user`. If the username is known but the PIN is not that user's PIN, print `Incorrect PIN`. If both are right, print `Welcome`, followed by the username.

Your program should produce these answers:

Username	PIN	Output
alice	1234	Welcome, alice
alice	9876	Incorrect PIN
bob	9876	Welcome, bob
bob	0000	Incorrect PIN
carol	1234	Unknown user

Hint: you should use a nested if statement

Problem 2 — Rock, Paper, Scissors

Write a program `rps.py` to play Rock, Paper, Scissors. The starter code chooses a random option with the follow two lines:

```
import random
bot = random.choice(['rock', 'paper', 'scissors'])
```

Use the prompt `player = input('Enter Choice')` to get your move, then print the winner - either bot or player. Paper beats rock, rock beats scissors, scissors beats paper. Try to beat the bot.

Hint: you should use `and` to check the values of both moves. Check for ties. If it isn't a tie, there are only three ways for the player to win: rock against scissors, scissors against paper, paper against rock. If it isn't a tie and the player doesn't win, then the bot wins by default.

Submission

Submit all three source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.

If you are not finished when the period ends, submit what you have. You may then complete it on your own and resubmit by Sunday, September 27 at 11:59 pm for up to 50% of the remaining credit.